General comments on EE2-6 Control Engineering paper 2019

Overall, the students have done well on the exam paper, scoring an exam average of 67%.

1. The students have done relatively well on this question, scoring approximately 79%. This is an op–amp modeling, including state–space, proportional controller design and Nyquist stability analysis question and is a variation on a past exam question and typical study group questions.
   (a) Tests understanding of transfer functions for simple circuits involving op–amps.
   (b) Tests obtaining differential equations from given transfer functions.
   (c) Tests obtaining state–space realizations from transfer functions.
   (d) Tests understanding of the use of the final value theorem in simple analysis.
   (e) Tests understanding of the use of the final value theorem in simple design for second order systems.
   (f) Tests understanding of the use of the final value theorem in more complex design.
   (g) Tests sketching Nyquist diagrams for simple systems.
   (h) Tests the use of the Nyquist theorem in simple proportional stability design.

2. This is a Nyquist diagram analysis and design question that involves a a third order system. It also tests understanding of the small gain theorem. The students did less well on this question scoring approximately 59%.
   (a) Tests the students' understanding of the use of the Routh–Hurwitz stability criterion in sketching Nyquist diagrams.
   (b) Requires an understanding of the use of the Routh–Hurwitz stability criterion to determine the stability margins and the cross–over frequency.
   (c) Qualitative analysis of the effects of PI compensation on the performance.
   (d) Tests the students' understanding of the relation between Nyquist analysis and the small gain theorem. Also, tests the student's understanding of the use of the small gain theorem in robust stability analysis.

3. This is a root–locus type design question for an unconventional feedback loop. It involves some manipulations to reduce the loop to a standard one for root–locus analysis and design. The students also did less well on this question, scoring approximately 59%.
   (a) Tests the students' understanding of the relation between design specifications and pole location.
   (b) Tests the students' understanding of using a parameter to partially achieve the specifications.
   (c) Tests the students' ability to use the root–locus to find the values of two parameters to achieve the design specifications.